

Sentient AI: Bridging Realities with the Fusion of DePINs and AI Agents

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Abstract

The convergence of Decentralized Physical Infrastructure Networks (DePINs) with AI Agents marks a new paradigm shift in redefining the landscape of infrastructure and intelligence. The seamless integration of AI agents with DePINs ensures real-time data processing, knowledge accumulation, and adaptive responses to changing environments and conditions. In this light paper, we highlight the benefits and potential use cases when connecting DePINs to AI agents and present QUICKSILVER, a connectivity layer to realize this vision and fuel growth and technological breakthroughs across a wide range of industry sectors.

1 Introduction

In the ever-evolving landscape of artificial intelligence, decentralized physical infrastructure networks (DePINs) are poised to play a pivotal role in the development of emerging AI agents that can truly understand and interact with the physical world. At its core, DePINs provide the essential framework for collecting real-world data in a decentralized manner. Those data can be used by AI agents to transcend their digital confines and engage with the physical realm. The convergence of DePINs and AI agents is essential because it addresses two fundamental requirements for advanced AI, namely *perception* and *action*.

For an AI agent to achieve a semblance of consciousness, it should be able to sense and interpret the physical world with a level of sophistication comparable to human perception. DePINs can help AI agents to achieve this goal by incentivizing community members to deploy a large number of smart devices across the physical environment. These networks of smart devices act as sensory organs for AI agents, providing continuous real-time data streams about their surroundings. The decentralized nature of these networks allows AI agents to accommodate dynamic and challenging environments.

Perception alone, however, is not sufficient for true sentience. The ability to act upon the physical world is equally crucial. DePINs provide AI agents with the means to interact with and manipulate their environment. This direct engagement with the physical world allows an AI agent not only to respond to its environment but also to refine its knowledge and understanding of cause and effect in the physical world, thereby establishing a positive feedback loop between perception and action.

By enabling perception and action capabilities, DePINs lay the groundwork for AI systems that can not only process real-world information but also experience and influence the tangible world around them. As we look to the future, the

integration of AI agents and DePINs is going to open a new era of human-machine interaction.

2 DePIN and AI Agents

DePINs serve as the bridge connecting the cognitive capabilities of AI agents with the tangible, physical world. This section gives an in-depth look at this integration.

2.1 DePINs as "Perception Organs"

As perception organs, DePINs play the following key roles for AI agents:

- **Sensors and Data Collection:** DePINs utilize networks filled with various physical devices — ranging from environmental sensors to sophisticated cameras — scattered globally. These devices accumulate a vast array of real-time environmental data such as temperature fluctuations, air quality indices, vehicular movements, and auditory signals.
- **Decentralized Data Accessibility:** In traditional settings, this data might be confined to specific silos, limiting operational capability. However, DePIN enables decentralized access to this wide array of information. AI agents can harness this data, updating their perception continuously and accommodating wide-reaching understandings of the environmental contexts.
- **Collective Perception:** The ability of multiple AI agents to concurrently tap into and utilize data from DePIN enhances shared situational awareness. This collective intelligence paradigm enables a more cohesive understanding and responses to dynamic real-world changes.

2.2 DePINs as "Hands and Feet"

As hands and feet, DePINs play the following key roles for AI agents:

- **Actuation and Execution Devices:** Alongside sensory inputs, DePIN embraces components capable of physically altering their environments — including robots, drones, and other smart devices equipped with actuators.
- **Decentralized Control Mechanisms:** This network allows AI agents to interface directly, permitting them to maneuver and manipulate physical settings through real-time interaction with connected devices.
- **Coordinated Actions:** Such interactions lead to operational syncretism where multiple agents engage seamlessly, promoting sophisticated behaviors and complex task handling through collective efforts.

3 Potential Use Cases

The symbiotic relationship between DePINs and AI agents has the great potential to revolutionize numerous industry sectors.

- **Autonomous Drone Delivery Network:** Imagine drones, equipped with sensors, operating under a unified DePIN framework. These could collaboratively access and process data regarding airspace, weather, and navigational obstacles. Such a network would enable drones, guided by AI, to optimize delivery pathways, evade potential hazards, and adjust to sudden aerial or environmental changes automatically.
- **Smart City Infrastructure:** Municipal systems could be revolutionized with AI agents monitoring integrated sensor data. Traffic management, energy distribution, pollution control, and emergency responses could be greatly enhanced through real-time data from a city-wide DePIN, leading to smarter, more responsive urban environments.
- **Decentralized Environmental Monitoring:** Deploying sensor networks in isolated or extreme environments can gather critical data regarding climate indicators, biodiversity, or ecological changes. AI agents analyzing this data can contribute significantly to environmental science, offering insights into climate trends, habitat conditions, or illegal activities such as deforestation.
- **Precision Agriculture:** In agriculture, robots and IoT devices interconnected through DePINs could track and analyze multiple factors like soil health, crop vitality, and micro-climate conditions. AI applications can process this data to fine-tune farming practices — optimizing resource allocation like water and fertilizers, thereby boosting yield while minimizing environmental impact.
- **Financial Predictions Agent:** Consider financial markets where an AI agent, via DePIN, accesses historical and real-time economic data to forecast trends or market movements. Such capabilities could transform financial analytics, offering predictions based on comprehensive, decentralized data streams.

4 Quicksilver: Connecting DePINs to AI Agents

To fully unlock the potential of AI agents, IoTeX has built Quicksilver [8], a connectivity layer between DePIN projects and AI Agents. As shown in Figure 1, Quicksilver creates a seamless interface between various DePINs and AI agents with the aid of the open source Eliza Operating System (ElizaOS) [9]. On the one hand, Quicksilver can help AI agents achieve perception capability by feeding real-time data from the physical world. On the other hand, AI agents can also orchestrate DePINs based on the agents' knowledge and decisions via Quicksilver. Built upon the IoTeX's robust and modular DePIN infrastructure, Quicksilver facilitates a fluid interaction between digital intelligence agents and the multitude of sensors and actuators in the physical world. BinoAI [4] is the first sentient DePIN AI agent on IoTeX, empowered by the ElizaOS. Figure 2 illustrates the responses from BinoAI when querying the weather information.

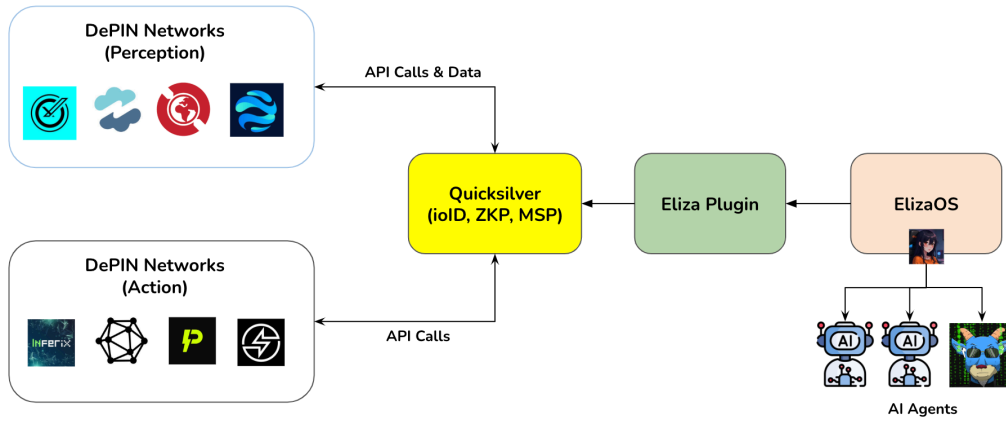


Figure 1: A High-Level Architecture of Quicksilver

```

npx ts-node example/demo_agent.ts
Enter your input (or "exit" to quit): What's the weather like in Rome?
LLM raw response: ``json
{
  "tool": "WeatherAPI",
  "tool_input": {
    "latitude": 41.9028,
    "longitude": 12.4964
  }
}
...
Agent Response:
The current weather in Rome is cloudy with a temperature of 13.2°C, but it feels
like 12.48°C. The humidity is at 73%, the pressure is 1019 hPa, and the wind sp
eed is 3.58 m/s.
----
Enter your input (or "exit" to quit): 
  
```

Figure 2: Responses from BinoAI when Querying the Weather Information

4.1 Building Blocks in Quicksilver

In a nutshell, Quicksilver is built upon a number of key DePIN infrastructure modules and can be viewed as a gigantic database of verified DePIN data from the entire ecosystem.

- **Identity Management.** Quicksilver relies on ioID [6; 5], a unified identity system for DePINs, managing device identities on-chain and off-chain. ioID lays down the trust foundation for DePIN data and also provides additional dual-mining opportunities for certain DePIN projects.
- **Verifiable Computation.** Quicksilver leverages W3bstream [3; 5] to realize verifiable computation on DePIN data. W3bstream utilizes advanced cryptographic techniques such as zero-knowledge proofs (ZKPs), trusted execution environments (TEEs), and secure multi-party computations (SMPC) to ensure integrity of data processing on Quicksilver nodes.

- **Data Storage.** Quicksilver integrates various centralized (e.g., cloud storage) and decentralized (e.g., IPFS [2]) storage solutions to provide short- and long-term storage for DePIN data.
- **Cryptoeconomic security.** Quicksilver utilizes Modular Security Pool (MSP) [7; 5] to realize cryptoeconomic security across the DePIN ecosystem. MSP acts as a pool gathering staking security from various established Layer 1 blockchains and distributing that security to new networks.

The combination of the above building blocks enables Quicksilver to provide verifiable DePIN data to AI agents, thereby significantly enhancing the security, reliability and robustness of AI agents. In particular, Quicksilver facilitates AI agents to interact with the physical world in real time, establishing a foundation for true autonomous decision making. This layered, integrated approach ensures a robust, scalable system poised to drive forward the potential for sentient AI applications.

4.2 Usage of IOTX Tokens

IOTX will be extensively utilized as utility tokens in different building blocks (i.e., ioID, W3bstream, MSP, etc.) of Quicksilver and the integration of Quicksilver with AI agents is going to bring numerous opportunities for users mining multiple DePIN-related tokens by joining this technological evolution.

4.3 Enhancement of DePIN-Oriented AI Agents

DePINs usually generate massive volumes of unstructured or semi-structured time-series data in practice, which poses additional challenges for AI agents providing accurate responses in a timely manner. Furthermore, DePINs cover a wide range of industry sectors and domain-specific knowledge also has a significant impact on the behavior of AI agents. To address these challenges, the large language module (LLM) that empowered DePIN-oriented AI agents should be further augmented by taking into consideration the unique characteristics of DePINs. Some recent research [1] has shed some light in this direction.

5 Conclusions and Future Outlook

The convergence of DePINs and AI agents heralds a new era of unprecedented synergy between artificial intelligence and the physical world and has the great potential to push the boundaries of technology innovation, industry revolution as well as enhancement of urban living. As DePINs continue to expand and evolve, we can anticipate AI agents that are not just more capable, but also more attuned to the nuances and complexities of our physical reality.

References

- [1] Tuo An, Yunjiao Zhou, Han Zou, and Jianfei Yang. Iot-llm: Enhancing real-world iot task reasoning with large language models, 2024.
- [2] Juan Benet. Ipfs - content addressed, versioned, p2p file system. <https://arxiv.org/pdf/1407.3561>, 2014.
- [3] IoTeX. W3bstream: A decentralized multi-prover network for depin verification. <https://docs.iotex.io/depin-infra-modules-dim/w3bstream-depin-verification>, 2023.
- [4] IoTeX. Introducing binoai: The first depin ai agent on iotex, powered by the eliza framework. <https://iotex.io/blog/binoai-the-first-depin-ai-agent-on-iotex/>, 2024.
- [5] IoTeX. Iotex 2.0: Depin for everyone! https://github.com/iotexproject/files/blob/main/publications/IoTeX_2_0_Whitepaper_1.1_EN.pdf, 2024.
- [6] IoTeX. ioID: A unified identity system for decentralized physical infrastructure networks. <https://docs.iotex.io/depin-infra-modules-dim/iodid-depin-identities>, 2024.
- [7] IoTeX. MSP: A unified trusted layer for depin infrastructure modules. <https://docs.iotex.io/depin-infra-modules-dim/msp-modular-security-pool>, 2024.
- [8] IoTeX. Quicksilver: Sentient ai framework. <https://github.com/iotexproject/quicksilver>, 2025.
- [9] Eliza Labs. Elizaos: The operating system for ai agents. <https://elizaos.ai/>, 2024.